Model selection for per-receptor odorant activity prediction: cross-validation versus nested cross-validation

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Submitted to Dr. Emil Sayilov • 2026-10-10

Question. The ENodorIM simulator predicts, for an arbitrary molecule, which of the human odorant receptors it activates. The predictor deployed on the site is one fixed configuration—an L_2-regularised logistic regression per receptor on Morgan fingerprint bits—reported with ordinary k-fold cross-validation. Two questions were conflated by that figure: *how good is this model*, and *how good is the procedure that chooses a model*. The second is the one that matters before tuning, and it requires the hyperparameter search to be nested inside the validation. This report answers both on identical folds.

Data. Human single-compound dose–response and screening experiments from M2OR, reduced to one consensus label per receptor–molecule pair, giving 654 molecules described by 2048-bit Morgan fingerprints (radius 2, chirality retained, because measured enantiomers activate receptors differently). 31 receptors have at least eight molecules on each side and are modelled; each has 70–319 tested molecules and 8–134 agonists, so every per-receptor data set is small and imbalanced.

Protocol. 5-fold stratified outer folds, repeated 3× with different shuffles; out-of-fold scores are pooled across a split before scoring, since a single fold can hold two positives. *Flat*: hyperparameters chosen on the outer folds and the winner's own score reported—what a tuning loop reports about itself. *Nested*: hyperparameters chosen by 3-fold inner cross-validation within each outer training fold, so selection is never graded on data it saw. Fingerprint-bit filtering is refitted per training fold. The difference between the two columns is the selection bias.

Model family	Grid	Flat	Nested	Optimism	Nested AP	Δ vs dep.	W/L
Logistic regression, $C = 0.1$ (deployed)	1	0.686	**0.686**	—	0.412	—	—
Random forest	4	0.707	0.685	+0.023	0.407	-0.001	14/17
Logistic regression, L2	6	0.707	0.677	+0.030	0.405	-0.009	8/23
Tanimoto k-nearest neighbours	3	0.703	0.674	+0.029	0.384	-0.012	11/20

Mean per-receptor ROC-AUC over 31 receptors. *Grid*: hyperparameter settings searched. *Optimism*: flat minus nested. *W/L*: receptors where the family's nested AUC beats or loses to the deployed model. No family differs from the deployed model at $p < 0.05$ (Wilcoxon signed-rank over paired receptors).

Findings. (i) No family beats the deployed configuration under nested cross-validation: it is itself the highest-scoring of the 4 at AUC 0.686, and the closest rival, Random forest, trails by -0.001; no family wins on a majority of receptors. (ii) Hyperparameter search inflates the flat-CV figure by +0.027 AUC on average—larger than the entire spread between the 4 families—so a tuned model reported by flat cross-validation would appear to beat the deployed one while being no better. (iii) Repeating the identical evaluation with a different fold shuffle moves a single receptor's AUC by ±0.021, again comparable to the between-model differences. The deployed similarity baseline, evaluated by hold-one-out over 654 reference molecules, ranks a true agonist first for 61% of molecules ($n = 138$).

Conclusion. On this data set model choice is not the binding constraint: 4 families spanning linear, kernel, ensemble and instance-based learning fall within the noise of one another, and the untuned deployed configuration is retained—it carries no selection bias to lose. Per-receptor performance is governed by the amount of measured data, not the algorithm, which is why the simulator reports each receptor's own cross-validated AUC and uses a learned score only above AUC 0.70 rather than quoting a single headline number. The next gain must come from more measured pairs, or from a model that shares one representation across receptors (a graph encoder with a receptor-sequence embedding), rather than from a wider hyperparameter search.

Reproducibility. `python -m scripts.pipeline.benchmark_models` (≈7 min, 31 receptors × 4 families) writes `data/processed/model_benchmark.json`, from which this report and the site's benchmark page are generated. Data: M2OR (DOI:10.1093/nar/gkad886). Code and results: `https://kammmran.github.io/ENodorIM/#/benchmark`.